Review Insight

Q.1 What is meant by private member method? [ICSE Model]

Ans. A member method declared to be private is used only within the class. Such method cannot be used outside the class visibility.

Q.2 Differentiate between private and protected visibility modifiers. [ICSE 2010]

Ans. Private members are only accessible within the class visibility. A protected member acts as a private member in the existing class but can be shared in derived class during inheritance.

Q.3 Name the keyword that:

(*i*) informs that an error has occurred in an input/output operation.

(*ii*) distinguishes between instant variables and class variables. [ICSE 2011]

Ans. (*i*) throws (*ii*) static

Q.4 Differentiate between public and private modifiers for the members of a class. [ICSE 2012]

Ans. Private modifier: A class member declared to be private has its access limited to the class block only. By no means, a private member can be applied outside the class block.
Public modifier: A class member that is declared to be public can be accessed inside as well as outside the class block. By default, a class member is declared to be public.

Q.5 In the program given below, state the name and the value of the

(*i*) method argument or argument variable

(*ii*) class variable

(*iii*) local variable

(*iv*) instance variable

```
class myClass
{
static int x=7;
int y=2;
public static void main(String args[])
 {
 myClass obj =new myClass();
 System.out.println(x);
 obj.sampleMethod(5);
 int a=6;
 System.out.println(a);
 }
void sampleMethod(int n)
 {
 System.out.println(n);
 System.out.println(y);
 }
}
```

Ans. (*i*) Method argument or argument variable n with value 5. [ICSE 2012]

(*ii*) Class variables: x with value 7.

(*iii*) Local variable: a with value 6.

(*iv*) Instance variable: y with value 2.

Q.6 Name the Java keyword that:

(*i*) indicates that a method has no return type.

(*ii*) stores the address of the currently calling object.

Ans. (*i*) void (*ii*) this [ICSE 2013]

Q.7 Identify the statements given below as assignment, increment, method invocation or object creation statements.

(i) Sytem.out.println("Java");
(ii) costPrice = 457.50;
(iii) Car hybrid = new Car();
(iv) petrolPrice++;

Ans. The statements under various categories are:

(i) System.out.println("Java"); : Method Invocation
(ii) costPrice = 457.50; : Assignment
(iii) Car hybrid = new Car( ); : Object Creation
(iv) petrolPrice++ : Increment

[ICSE 2014]

Q.8 List the variables that are composite data types from those given below.

(i) static int x;
(ii) arr[i] = 10;
(iii) obj.display();
(iv) Boolean b;
(v) private char chr;
(vi) String str;

Ans. The variables those can be categorized under composite data types are as follows:

(i) arr[i] = 10;
(ii) obj.display( );

[ICSE 2014]

Q.9 Consider the following class:

```
public class MyClass
{
public static int x-3, y=4;
public int a=2, b=3;
}
```

(i) Name the variables for which each object of the class will have its own distinct copy.
(ii) Name the variables that are common to all objects of the class. [ICSE 2014]

Ans. (i) The variables for which each object of the class will have its own distinct copy are a and b.

(ii) The variables that are common to all the objects of the class are x and y.

Q.10 The access specifier that gives the most accessibility is ..... and the least accessibility is ..... . [ICSE 2015]

Ans. public, private

Q.11 Name the keyword which is used to resolve the conflict between method parameter and instance variables/fields. [ICSE 2015]

Ans. this keyword

Q.12 Name any two types of access specifiers. [ICSE 2016]

Ans. The two type of access specifiers are:

(i) public (ii) private

Q.13 Case-Study based question:

A class is a blueprint or prototype or template to create similar objects. Each object of the class contains the characteristics and behaviour described within the class. The characteristics and behaviour of a real world object resemble data and functions (methods) of a software object. You can use a data field that is common for all the objects of the class. The data members of a class can be declared such that their scope can be limited to the scope of the class or entire program. Java language also provides the keyword to distinguish the current object from the object passed to the method. Based on the above discussion, answer the following questions:

(a) An object is also called ..... of a class.
(i) substance (ii) instance (iii) resonance (iv) None

(b) The data members declared within a class is also termed as:
(i) instance variables (ii) local variables
(iii) global variables (iv) temporary variables

(c) What type of data is accessible only within the scope of a class?
(i) public (ii) protected (iii) private (iv) secure

(d) Name the keyword which distinguishes current object and the object passed to the method.
(i) new (ii) final (iii) static (iv) this

Ans. (a) (ii) instance (b) (i) instance variables
(c) (iii) private (d) (iv) this

EXERCISES

I. Multiple Choice Questions:

1. Which keyword makes class members accessible outside the class in which they are declared?
(a) Private (b) Protected (c) Public (d) Hidden

2. Find the access specifier which prohibits a class member from being used outside a class:
(a) Private (b) Public (c) Protected (d) None

3. A class object is also known as:
(a) Identifier (b) Instance variable (c) Specifier (d) Modifier

4. Which of the following statements is the most appropriate for the private members?
(a) They are visible out of the class in which they are defined.
(b) They can be used in the sub-classes.
(c) They are only visible in the class in which they are declared.
(d) None of the above.

5. Which of the following keywords are used to control access to a class member?
(a) Default (b) Abstraction (c) Protected (d) Interface

6. Which of the members can be accessed globally?
(a) Private (b) Public (c) Protected (d) All of the above

7. The maximum number of objects of a class can be created as:
(a) 1 (b) 2
(c) On the user's choice (d) Number of variables

8. A class contains:
(a) Attributes and methods. (b) A number of object of same types.
(c) Data and member function. (d) All of the above

9. Which of the following features is not the principle of OOP?
(a) Encapsulation (b) Transparency
(c) Inheritance (d) Polymorphism

10. A package is a:
(a) collection of data. (b) collection of functions.
(c) collection of classes. (d) a nested class.

II. Fill in the blanks with appropriate words:

1. Primitive data types are also called as ..... data types.

2. A user defined data type can be created by using a/an ..... .

3. ..... keyword represents the current object in the member method.

4. ..... members are accessible from anywhere in the program.

5. If no access specifier is mentioned then ..... specifier is referred by default.

6. ..... members are accessible only within the same class.

7. ..... members are accessible in its own class as well as in a sub class.

III. Case-Study based question:

Given below is a class based program to accept name and price of an article and find the amount after 12% discount if the price exceeds 10,000 otherwise, discount is nil. There are some places in the program left blank marked with ?1?, ?2?, ?3? and ?4? to be filled with appropriate keyword/expression.

```
class Discount
{
  int pr; double d, amt; String nm;
  Scanner ob = ....?1?.....Scanner(System.in);
void input( )
{
  System.out.println("Enter customer's name:");
  nm=....?2?.... ;
  System.out.println("Enter price of the article:");
  pr = ob.nextInt( );
}
void calculate()
{
  if (...?3?...)
    d= .....?4?.....;
  else
    d=0;
  amt = pr – d;
}
void print()
{
System.out.println("Name ="+nm);
System.out.println("Amount to be paid="+amt);
}
```

Based on the above discussion, answer the following questions:

(*a*) What will be keyword/expression filled in place of ?1?

(*b*) What will be keyword/expression filled in place of ?2?

(*c*) What will be keyword/expression filled in place of ?3?

(*d*) What will be keyword/expression filled in place of ?4?

IV. Answer the following questions:

1. Why is a class known as composite data type? [ICSE 2009]
2. Name the types of data used in a class. [ICSE 2006]
3. What is the purpose of the new operator?
4. Can a class be referred to as user defined data type? Comment.

5. What is public access of a class?
6. How are private members of a class different from public members?
7. Mention any two attributes required for class declaration.
8. Explain instance variables. Give an example. [ICSE 2008]
9. Explain any two types of access specifiers. [ICSE 2008]
10. What is meant by private visibility of a method? [ICSE 2008] [ICSE 2006]

V. Answer the questions given below (Long answer type):

1. 'Object is an Instance of a class.' Explain this statement.
2. Differentiate between built-in data types and user defined data types.
3. Which of the following declarations are illegal and why?
   (*a*) class abc(...)
   (*b*) public class NumberOfDaysWorked(...)
   (*c*) private int x;
   (*d*) private class abc(...)
   (*e*) default key getkey(...)
4. Why can't every class be termed as user defined data type?
5. Differentiate between static data members and non-static data members. [ICSE 2010]
6. Differentiate between private and protected visibility modifiers. [ICSE 2010]
7. Differentiate between instance variable and class variable. [ICSE 2011]

VI. Unsolved Programs based on Class:

1. Define a class Calculate to accept two numbers as instance variables. Use the following member methods for the given purposes:

   Class name : Calculate
   Data members : int a, int b
   Member methods:
   void inputdata() : to input both the values
   void calculate() : to find sum and difference
   void outputdata() : to print sum and difference of both the numbers.
   Use a main method to call the functions.

2. Define a class Triplet with the following specifications:
   Class name : Triplet
   Data members : int a, int b, int c
   Member methods:
   void getdata() : to accept three numbers
   void findprint() : to check and display whether the numbers are Pythagorean Triplets or not.

3. Define a class Employee having the following description:
   Class name : Employee
   Data members/Instance variables
   int pan : to store personal account number
   String name : to store name
   double taxincome : to store annual taxable income
   double tax : to store tax that is calculated
   Member functions:
   void input( ) : Store the pan number, name, taxable income
   void cal( ) : Calculate tax on taxable income.
   void display( ) : Output details of an employee
   Calculate tax based on the given conditions and display the output as per the given format.

Total Annual Taxable Income	Tax Rate
Up to ₹2,50,000	No tax
From ₹2,50,001 to ₹5,00,000	10% of the income exceeding ₹2,50,000
From ₹5,00,001 to ₹10,00,000	₹30,000 + 20% of the income exceeding ₹5,00,000
Above ₹10,00,000	₹50,000 + 30% of the income exceeding ₹10,00,000

Output:

Pan Number	Name	Tax-Income	Tax
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....
.....	.....	.....	.....

[ICSE 2008]

4. Define a class Discount having the following description:

Class name : Discount

Data members

int cost : to store the price of an article
String name : to store the customer's name
double dc : to store the discount
double amt : to store the amount to be paid

Member methods:

void input( ) : Stores the cost of the article and name of the customer
void cal( ) : Calculates the discount and amount to be paid
void display( ) : Displays the name of the customer, cost, discount and amount to be paid

Write a program to compute the discount according to the given conditions and display the output as per the given format.

List Price	Rate of discount
Up to ₹5,000	No discount
From ₹5,001 to ₹10,000	10% on the list price
From ₹10,001 to ₹15,000	15% on the list price
Above ₹15,000	20% on the list price

Output:

Name of the customer	Discount	Amount to be paid
.....	.....	.....
.....	.....	.....
.....	.....	.....

5. Define a class Telephone having the following description:

Class name : Telephone

Data members

int prv,pre : to store the previous and present meter readings
int call : to store the calls made (i.e. pre – prv)
String name : to store name of the consumer
double amt : to store the amount
double total : to store the total amount to be paid

Member functions:

void input( ) : Stores the previous reading, present reading and name of the consumer

void cal( ) : Calculates the amount and total amount to be paid

void display( ) : Displays the name of the consumer, calls made, amount and total amount to be paid

Write a program to compute the monthly bill to be paid according to the given conditions and display the output as per the given format.

Calls made	Rate
Up to 100 calls	No charge
For the next 100 calls	90 paise per calls
For the next 200 calls	80 paise per calls
More than 400 calls	70 paise per calls

However, every consumer has to pay ₹180 per month as monthly rent for availing the service.

Output: **Name of the customer** **Calls made** **Amount to be paid**

....

....

6. Define a class Interest having the following description:

Class name : Interest

Data members :

int p : to store principal (sum)

int r : to store rate

int t : to store time

double interest : to store the interest to be paid

double amt : to store the amount to be paid

Member methods:

void input( ) : Stores the principal, rate, time

void cal( ) : Calculates the interest and amount to be paid

void display( ) : Displays the principal, interest and amount to be paid

Write a program to compute the interest according to the given conditions and display the output.

Time	Rate of interest
For 1 year	6.5%
For 2 years	7.5%
For 3 years	8.5%
For 4 years or more	9.5%

(**Note**: Time to be taken only in whole years)

7. Define a class Library having the following description:

Class name : Library

Data members

String name : to store name of the book

int price : to store the printed price of the book

int day : to store the number of days for which fine is to be paid

double fine : to store the fine to be paid

Member methods:

void input( ) : To accept the name of the book and printed price of the book

void cal( ) : Calculates the fine to be paid

void display( ) : Displays the name of the book and fine to be paid

Write a program to compute the fine according to the given conditions and display the fine to be paid.

Days	Fine
First seven days	25 paise per day
Eight to fifteen days	40 paise per day
Sixteen to thirty days	60 paise per day
More than thirty days	80 paise per day

8. Bank charges interest for the vehicle loan as given below:

Number of years	Rate of interest
Up to 5 years	15%
More than 5 and up to 10 years	12%
Above 10 years	10%

Write a program to model a class with the specifications given below:

Class name : Loan

Data members/Instance variables

int time : Time for which loan is sanctioned
double principal : Amount sanctioned
double rate : Rate of interest
double interest : To store the interest
double amt : Amount to pay after given time

Member Methods:

void getdata( ) : to accept principal and time.
void calculate( ) : to find interest and amount.
Interest = (Principal*Rate*Time)/100
Amount = Principal + Interest
void display( ) : to display interest and amount.

9. Hero Honda has increased the cost of its vehicles as per the type of the engine using the following criteria:

Type of Engine	Rate of increment
2 stroke	10% of the cost
4 stroke	12% of the cost

Write a program by using a class to find the new cost as per the given specifications:

Class name : Honda

Data members/Instance variables

int type : to accept type of engine 2 stroke or 4 stroke
int cost : to accept previous cost

Member Methods:

void gettype( ) : to accept the type of engine and previous cost
void find( ) : to find the new cost as per the criteria given above
void printcost( ) : to print the type and new cost of the vehicle

10. Define a class called 'Mobike' with the following specifications:

Class name : Mobike

Instance variables/ Data members:

int bno : to store the bike number
int phno : to store the phone number of the customer
String name : to store the name of the customer
int days : to store the number of days the bike is taken on rent
int charge : to calculate and store the rental charge

Member methods:

void input( ) : to input and store the details of the customer

void compute( ) : to compute the rental charge

The rent for a mobike is charged on the following basis:

For first five days : ₹ 500 per day

For next five days : ₹ 400 per day

Rest of the days : ₹ 200 per day

void display( ) : to display the details in the following format: [ICSE 2011]

Bike No.	Phone No.	Name	No. of days	Charge
xxxxxx	xxxxxxxx	xxxxxxxxx	xxx	xxxxxx

11. Write a program using a class with the following specifications:

Class name : Caseconvert

Data members : String str

Member functions:

void getstr() : to accept a string

void convert() : to obtain a string after converting each uppercase letter into lowercase and vice versa.

void display() : to print the converted string.

12. Write a program by using a class with the following specifications:

Class name : Vowel

Data members : string s; int c(to count vowels)

Member functions :

void getstr() : to accept a string

void getvowel() : to count the number of vowels

void display() : to print the number of vowels.

13. A bookseller maintains record of books belonging to the various publishers. He uses a class with the specifications given below:

Class name : Stock

Data members/Instance variables

String title : Contains title of the book.

String author : Contains author name.

String pub : Contains publisher's name.

int noc : Number of copies.

Member Methods:

void getdata( ) : to accept title, author, publisher's name and the number of copies.

void purchase(int t, String a, String p, int n):

to check the existence of the book in the stock by comparing total, author's and publisher's name. Also check whether noc >n or not. If yes, maintain the balance as noc-n, otherwise display book is not available or stock is under flowing.

Write a program to perform the task given above.

14. Write a program by using class with the following specifications:

Class name : Characters

Data members/Instance variables

str : String

Member Methods:

void input (String st) : to assign st to str

void check_print( ) : to check and print the following:

(*i*) number of letters (*ii*) number of digits

(*iii*) number of uppercase characters (*iv*) number of lowercase characters

(*v*) number of special characters

15. Define a class Student with the following specifications:

Class name : **Student**

Instance variables/Data Members:

String name : to store the name of the student

int eng : to store marks in English

int hn : to store marks in Hindi

int mts : to store marks in Maths

double total : to store total marks

double avg : to store average marks

Member methods:

void accept( ) : to input marks in English, Hindi and Maths

void compute( ) : to calculate total marks and average of 3 subjects

void display( ) : to show all the details viz. name, marks, total and average

Write a program to create an object and invoke the above methods.

16. Define a class called ParkingLot with the following description:

Class name : ParkingLot

Instance variables/Data members:

int vno : to store the vehicle number

int hours : to store the number of hours the vehicle is parked in the parking lot

double bill : to store the bill amount

Member methods:

void input( ) : to input the vno and hours

void calculate( ) : to compute the parking charge at the rate ₹ 3 for the first hour or the part thereof and ₹ 1.50 for each additional hour or part thereof.

void display( ) : to display the detail

Write a main method to create an object of the class and call the above methods. [ICSE 2015]

17. Design a class RailwayTicket with following description:

Class name : RailwayTicket

Instance variables/Data members:

String name : to store the name of the customer

String coach : to store the type of coach customer wants to travel

long mob no : to store customer's mobile number

int amt : to store basic amount of ticket

int totalamt : to store the amount to be paid after updating the original amount

Member methods:

void accept( ) : to take input for name, coach, mobile number and amount.

void update( ) : to update the amount as per the coach selected (extra amount to be added in the amount as follows)

Type of Coaches	Amount
First_AC	₹700
Second_AC	₹500
Third_AC	₹250
Sleeper	None

void display( ) : to display all details of a customer such as name, coach, total amount and mobile number.

Write a main method to create an object of the class and call the above member methods. [ICSE 2018]